ZCAS University

BAC4301 CORPORATE FINANCE

SAMPLE QUESTION PAPER

TIME ALLOWED: THREE HOURS

(5 minutes to read through the paper)

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: **Answer Three (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: COMPULSORY QUESTION

QUESTION 1:

ETM Co is considering two different projects. Project A is a local project with all cash flows in kwacha while project B is a foreign project with cash flows in US dollars.

Project A:

This involves producing a new product using a new equipment which will be acquired immediately at a cost of K530,000. The asset will last for 4 years after which the scrap value will be K40,000. Forecast sales units, selling prices, and production costs in money terms due to the new product are as follows;

Year	**1**	**2**	**3**	**4**
Sales units	30,000	25,000	20,000	10,000
Selling price (K/unit)	14	15	16	18
Variable cost (K/unit)	4	6	6	10

The company tax is 30% payable one year in arrears. Capital allowances are available at a 20% reducing balance basis. The after-tax money cost of capital is 10%.

REQUIRED:

Calculate the net present value (NPV) of the project and comment on the results. **[18 marks]**

Project B:

TC plc has a five-year project in the US with cashflows denominated in dollars as follows;

Year	**0**	**1**	**2**	**3**	**4**	**5**
Cash flows **$**	(12,000)	2,500	4,500	4,000	5,000.	2,000

The ungeared cost of equity for the dollar cash flows is 8% while the spot exchange rate kwacha to US$ is K17.5 to US$1.

REQUIRED:

i. Calculate the base case NPV in dollars and comment on the results. **[7 marks]**

ii. Assuming that TC plc proposes to use $9,000 of 5% debt to finance the project. The debt is repayable in equal annual instalments of $1,800 while interest is paid on a reducing balance basis. Company tax is at 30% per year.

Calculate the APV of the project in dollars and in kwacha?. [11 marks]

iii. Briefly explain the steps involved in appraising a foreign project based on the spot exchange rate. **[4 marks]**

[TOTAL: 40 MARKS]

SECTION B: ATTEMPT <u>ANY THREE</u> QUESTIONS IN THIS SECTION.

<u>QUESTION TWO</u>

PS plc is evaluating a seven-year project in the renewable energy sector. The initial outlay is K5.8 million and the expected cash flows originating from the project are K700, 000 for the first year, then increasing by K250, 000 each year in years 2 and 3, then remaining at year 3 level for year 4 and 5 before reducing to K1,000,000 per year for year 6 and 7.

The project will be financed purely by equity. PS has an equity beta of 0.95 while the yield on treasury bills is 7%. The expected market return is 11 percent.

<u>REQUIRED:</u>

a) Calculate the cost of equity for PS plc. **[2 marks]**

b) Calculate the MIRR for the project and comment on the viability of the project. **[12 marks]**

c) Briefly discuss any THREE practical factors that may influence a firm's decision to use debt finance in its capital structure. **[6 marks]**

[Total: 20 marks]

<u>QUESTION THREE</u>

Management of ERS ltd are in a process of putting up a comprehensive risk management policy of the business. Of concerned is the interest rate risk and the exchange rate exposure associated with firm's active cross border trading. Currently the firm has the following projected cash movement from its business operations:

- **Expected to payment of $16,500 today**
- **Expected payment of $21,000 in two months' time.**

Exchanges rate K per $:

Spot exchange rate	17.102	17.155
1 month's forward rate	17.540	17.598
2 months forward rate	17.872	17.901

REQUIRED:

a) Calculate the kwacha cost of the $16,500 payable today. **[3marks]**

b) Determine the exchange gains or losses associated with using a forward contract as compared to the current exchange rates on the $21,000 payable in two months' time**.** **[6 marks]**

c) Briefly explain how the forward rate agreement (FRA) is used as a strategy to hedge interest rate risk. **[6marks]**

d) Define what an option is and differentiate a put option from call option. **[5 marks]**

[Total 20 marks]

QUESTION FOUR

MBS has the following capital structure 100m (K0.60) ordinary shares K60m, 8% K100 irredeemable preference shares K40m and K20m of B-rated 5-year bonds. A similar 5-year government bond has a yield of 7%.

The ordinary shares are currently trading at K1.20 ex-div per share while preference shares are quoted at K80 ex-div per K100 nominal value. MBS has just paid dividend of 9 ngwee per share. The dividend growth rate is constant at 4% per annum. The company faces a tax rate of 33%. The beta value of the firm's ordinary shares is 0.95.

REQUIRED:

a) Calculate WACC for MBS using market value weights. **[14 marks]**

b) Calculate and comment on the firms gearing ratio using the book values (debt to equity ratio). The industry average is 40**%.** **[4 marks]**

c) Outline any TWO disadvantages of excessive borrowing. **[2 marks]**

[Total: 20 Marks]

QUESTION FIVE

ETM plc is considering acquiring 100% shares in MKL plc through a share-for-share exchange by issuing 20m shares in ETM for the 40m shares in MKL.

	ETM plc	**MKL Plc**
Number of shares	**150m**	**40m**
Total earnings	**K12m**	**K6m**
P/E ratio	**37.5**	**5**
Earnings per share	**K0.08**	**K0.15**

REQUIRED:

a) For EACH of the above firms (ETM and MKL) calculate the current share price, and the market value of each company using the P/E ratio basis. **[8 marks]**

b) Determine the post-acquisition EPS if the synergetic earnings are K16m. **[4 marks]**

c) Calculate the share price of ETM post-acquisition if the bidder's P/E increases by 2.5 times. **[2 marks]**

d) Briefly outline the main features associated with the following defense tactics in case of a hostile takeover**;**

i. Poison pill. **[3marks]**

ii. White knight**.** **[3 marks]**

[Total 20 marks]

END OF EXAMINATION

Credit Spread by type of bond and maturity Corporate spreads for Industries (in basis points) 1 basis point = 0.01%							
Rating	**1 year**	**2 year**	**3 year**	**5 year**	**7 year**	**10 year**	**30 year**
AAA	**5**	**10**	**15**	**20**	**25**	**33**	**60**
AA	**15**	**25**	**30**	**35**	**44**	**52**	**71**
A	**35**	**44**	**55**	**60**	**65**	**73**	**90**
BBB	**60**	**75**	**100**	**105**	**112**	**122**	**143**
BB	**140**	**180**	**210**	**205**	**210**	**250**	**300**
B	**215**	**220**	**260**	**300**	**315**	**350**	**450**
CCC	**1,125**	**1,225**	**1,250**	**1,200**	**1,200**	**1,275**	**1,400**

FORMULAE SHEET

1. **Discount factor =** $(\frac{1}{1+r})^{n}$ **Annuity factor =** $\frac{1}{r}-\frac{1}{(1+r)^{n}}$

2. **IRR (YTM) =** $A + \left[\frac{}{a-b} \times (B - A)\right]_{a}$

3. **MIRR =** $\left[\sqrt[N]{\frac{FVCF(inflows)}{PVCF\ (Outflows)}}\right] - 1$

4. **CAPM:** $r_i = r_f + \beta_i \{r_m - r_f\}$

5. **g =** $\sqrt[n]{\left(\frac{Latest\ dividend}{Earliest\ dividend}\right)} - 1$

6. **WACC =** $re\left(\frac{Ve}{Ve+Vd}\right) + rd(1 - t)\left(\frac{Ve}{Ve+Vd}\right)$

7. **Corporate yield = risk free rate + credit spread**

8. $K_e = \frac{do\ (1 + g)}{P_o} + g$ **or** $D1/Po + g$

9. **Beta x =** $\frac{Correlation\ x\ with\ market * Std\ dev\ of\ return\ x}{std\ dev\ of\ market\ return}$